

## Submission LINALG1-PS1-SUB02

### Question LINALG1-PS1-Q01

work / setup:

1.  $\det = 6(5) - 1(4) = 26$

2. Eliminate  $y$ , yielding  $x=1$ ; substitute  $\Rightarrow y=2$

crossed-out arithmetic: [smudged]

FINAL (maybe):  $x=1$ ,  $y=2$  plus 1

### Question LINALG1-PS1-Q02

work / setup:

1.  $\det(A) = 2 \cdot 3 - 2 \cdot 2 = 2$

crossed-out arithmetic: [smudged]

FINAL (maybe):  $\det=2$ ; invertible plus 1

### Question LINALG1-PS1-Q03

work / setup:

1.  $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$

2. Matrix  $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$

crossed-out arithmetic: [smudged]

FINAL (maybe):  $T(1, -1) = (-1, 4)$ ; matrix  $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ ; linear plus 1

### Question LINALG1-PS1-Q04

work / setup:

1. Set  $c \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ .

crossed-out arithmetic: [smudged]

FINAL (maybe): only trivial combination;  $\{(1,0), (0,1)\}$  is linearly independent plus 1

### Question LINALG1-PS1-Q05

work / setup:

1.  $\text{Null}(A)$  is the set of vectors sent to the zero vector.

crossed-out arithmetic: [smudged]

FINAL (maybe): the vectors mapped to zero; exactly the solution space of  $Ax=0$  plus 1

### Question LINALG1-PS1-Q06

work / setup:

1.  $\det = 5(3) - 2(3) = 9$

2. Eliminate  $y$ , yielding  $x=1$ ; substitute  $\Rightarrow y=2$

crossed-out arithmetic: [smudged]

FINAL (maybe):  $x=1$ ,  $y=2$  plus 1